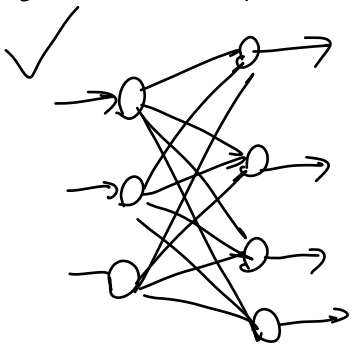
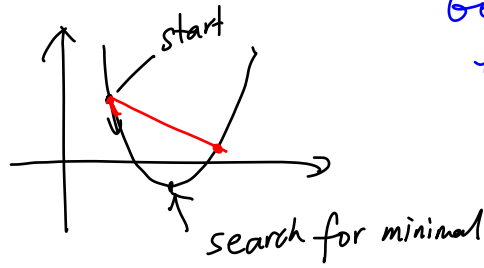


Lecture 5. Vector Calculus and Optimization: Differential and Gradient



$\vec{y} = \underline{W}\vec{x} + \vec{b}$ we would like to obtain W , which is trainable parameter such that $\|y - \hat{y}\|$ is minimal.
 $\Rightarrow$ how to search?



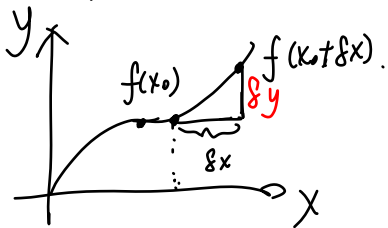
General idea: along directions that gradient decaying.

- ① how to compute gradient?
- ② how to go along the direction of gradient decent?

1. Differentiation of univariate functions.

(1) Basic concept: derivative.

Def: Consider a univariate function $y = f(x)$, $x, y \in \mathbb{R}$, $\mathbb{R} \rightarrow \mathbb{R}$



Different Quotient.

$$\frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x} \xrightarrow{\delta x \rightarrow 0}$$

Derivative

$$\frac{df}{dx} := \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

Some common rule for derivative.

① $f(x) = x^n$, $f'(x) = nx^{n-1}$; ② $f(x) = \sin x$, $f'(x) = \cos x$

③ $f(x) = \cos x$, $f'(x) = -\sin x$; ④ $f(x) = e^x$, $f'(x) = e^x$

⑤ $f(x) = \ln x = \log_e x$, $f'(x) = \frac{1}{x}$

(2) Chain rule for derivative.

① $(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$

② $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

③ $(g(f(x)))' = (g \circ f)'(x) = g'(f(x)) \cdot f'(x)$

Example: $h(x) = (2x+1)^4$

Let $f(x) = 2x+1$, $g(f) = f^4$

$f'(x) = 2$, $g'(f) = 4f^3 \Rightarrow h'(x) = g'(f(x)) \cdot f'(x)$

$= 4f^3 \cdot 2 = 4(2x+1)^3 \cdot 2$
 $= 8(2x+1)^3$

2. Partial differential & gradients:

Consider a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (requires derivative to function of several variables).

1) Def. Partial Derivative:

For a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$ of n variable $x_1, \dots, x_n$ the partial derivatives are.

$$\textcircled{2} \quad \left[\frac{\partial f}{\partial x_1} \right] = \lim_{h \rightarrow 0} \frac{f(x_1+h, x_2, x_3, \dots, x_n) - f(\vec{x})}{h}$$

$$\vdots$$

$$\left[\frac{\partial f}{\partial x_n} \right] = \lim_{h \rightarrow 0} \frac{f(x_1, x_2, \dots, x_n+h) - f(\vec{x})}{h}$$

$$\Rightarrow \nabla_x f = \frac{df}{d\vec{x}} = \left[\frac{\partial f(\vec{x})}{\partial x_1} \quad \frac{\partial f(\vec{x})}{\partial x_2} \quad \dots \quad \frac{\partial f(\vec{x})}{\partial x_n} \right] \in \mathbb{R}^{1 \times n} \text{ (row vector)}$$

$$\vec{x} = [x_1 \dots x_n]^T$$

(2) Chain rule for differentiation

$$\textcircled{1} \quad \frac{\partial}{\partial \vec{x}} (f(\vec{x}) \cdot g(\vec{x})) = \frac{\partial f}{\partial \vec{x}} g(\vec{x}) + f(\vec{x}) \cdot \frac{\partial g}{\partial \vec{x}}$$

$$\textcircled{4} \quad \frac{\partial}{\partial x} \left(\frac{f}{g} \right) = \frac{\frac{\partial f}{\partial x} \cdot g - \frac{\partial g}{\partial x} \cdot f}{g^2}$$

$$\textcircled{2} \quad \frac{\partial}{\partial \vec{x}} (f(\vec{x}) + g(\vec{x})) = \frac{\partial f}{\partial \vec{x}} + \frac{\partial g}{\partial \vec{x}}$$

Summary: Similar to 1-D case.

Derivative $\rightarrow$ partial derivative

$$\textcircled{3} \quad \frac{\partial}{\partial \vec{x}} (g \circ f)(\vec{x}) = \frac{\partial g}{\partial f} \cdot \frac{\partial f}{\partial \vec{x}}$$

Example: Given $f(x, y) = (x + 2y^3)^2$, compute $\nabla_x f$.

$$g(x, y) = x + 2y^3 \quad f(g) = \underline{g^2}$$

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial x} = 2g \cdot \frac{\partial}{\partial x} (x + 2y^3)$$

$$= 2(x + 2y^3)$$

$$\frac{\partial f(x, y)}{\partial y} = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial y} = 2g \cdot \frac{\partial}{\partial y} (x + 2y^3)$$

$$= 2 \cdot (x + 2y^3) \cdot 6y^2 = 12(x + 2y^3)y^2$$

$$\underline{\nabla_x} = [2(x + 2y^3) \quad 12(x + 2y^3) \cdot y^2]$$

why row vector?

case 1: Consider function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ of variable x_1, x_2
 $x_1(t), x_2(t): \mathbb{R} \rightarrow \mathbb{R}$

$$\rightarrow \frac{df}{dt} = \frac{\partial f}{\partial x_1} \cdot \frac{\overset{dx_1}{dt}}{\frac{dt}{dt}} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{dt} = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right] \begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \end{bmatrix}$$

case 2: $x_1(s, t), x_2(s, t): \mathbb{R}^2 \rightarrow \mathbb{R}$, let $\vec{\theta} = [s, t]$

$$\text{since } \frac{\partial f}{\partial s} = \frac{\partial f}{\partial x_1} \cdot \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \cdot \frac{\partial x_2}{\partial s}$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x_1} \cdot \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \cdot \frac{\partial x_2}{\partial t}$$

$$\Rightarrow \frac{df}{d\vec{\theta}} = \frac{\partial f}{\partial \vec{x}} \cdot \frac{\partial \vec{x}}{\partial \vec{\theta}} = \left[\frac{\partial f}{\partial s} \quad \frac{\partial f}{\partial t} \right]$$

$$\text{Reason: row vector representation is more natural without requiring transposing.} \quad = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \right] \cdot \begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}$$

total derivative. $df_a = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) \cdot dx_i$

So far.

① $f: \mathbb{R}^D \rightarrow \mathbb{R}$: the gradient is a $1 \times D$ row vector.

② $f: \mathbb{R} \rightarrow \mathbb{R}^E$: the gradient is an $E \times 1$ column vector.

③ $f: \mathbb{R}^D \rightarrow \mathbb{R}^E$: the gradient is an $E \times D$ matrix.

3. Gradients of vector-valued functions

Considering function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, with $\vec{x} = [x_1, \dots, x_n]^T \in \mathbb{R}^n$.

The corresponding vector of functions:

$$\vec{f}[\vec{x}] = \begin{bmatrix} f_1(\vec{x}) \\ \vdots \\ f_n(\vec{x}) \end{bmatrix} \in \mathbb{R}^m, \text{ with } \vec{f} = [f_1, \dots, f_m]^T \text{ s.t. } f_i: \mathbb{R}^n \rightarrow \mathbb{R}, i=1, \dots, m$$

$$\frac{\partial \vec{f}}{\partial \vec{x}} = \left[\frac{\partial f_1(\vec{x})}{\partial x_1} \quad \dots \quad \frac{\partial f_m(\vec{x})}{\partial x_n} \right]$$

$$= \begin{bmatrix} \frac{\partial f_1(\vec{x})}{\partial x_1} & \dots & \frac{\partial f_1(\vec{x})}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m(\vec{x})}{\partial x_1} & \dots & \frac{\partial f_m(\vec{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

← collection of all first-order partial derivative of a vector function $\vec{f}$
 $\Rightarrow$ Jacobian matrix.

Exercise: Given

$$\vec{f}(\vec{x}) = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix} = \begin{bmatrix} x_1^2 x_2 + x_1 x_2^3 \\ x_1 x_2 + 3x_2 + 2x_1 x_2^4 \end{bmatrix} \in \mathbb{R}^{2 \times 1}, \text{ compute } \nabla_{\vec{x}} \vec{f}$$

with $\vec{x} = [x_1, x_2]^T$

$$\nabla_{\vec{x}} \vec{f} = \begin{bmatrix} \frac{\partial f_1(x_1, x_2)}{\partial x_1} & \frac{\partial f_1(x_1, x_2)}{\partial x_2} \\ \frac{\partial f_2(x_1, x_2)}{\partial x_1} & \frac{\partial f_2(x_1, x_2)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2x_2 x_1 + x_2^3 & x_1^2 + 3x_1 x_2^2 \\ x_2 + 2x_1^4 & x_1 + 3 + 8x_1 x_2^3 \end{bmatrix}$$

$$\frac{\partial f_1(x_1, x_2)}{\partial x_1} = 2x_2 x_1 + x_2^3$$

$$\frac{\partial f_1(x_1, x_2)}{\partial x_2} = x_1^2 + 3x_1 x_2^2$$

$$\frac{\partial f_2(x_1, x_2)}{\partial x_1} = x_2 + 2x_1^4$$

$$\frac{\partial f_2(x_1, x_2)}{\partial x_2} = x_1 + 3 + 8x_1 x_2^3$$

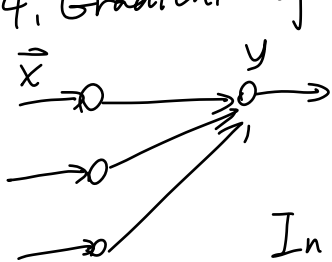
Special case: linear mapping.

Given $\vec{f}(\vec{x}) = A\vec{x}$, $A \in \mathbb{R}^{M \times N}$, $\vec{f}(\vec{x}) \in \mathbb{R}^{M \times 1}$, $\vec{x} \in \mathbb{R}^{N \times 1}$, compute

$$\nabla_{\vec{x}} \vec{f} = \frac{d\vec{f}}{d\vec{x}} \quad \text{Note: } \vec{f}: \mathbb{R}^N \rightarrow \mathbb{R}^M \Rightarrow \frac{d\vec{f}}{d\vec{x}} \in \mathbb{R}^{M \times N} = \underline{A}$$

$$f_i(\vec{x}) = \sum_{j=1}^N A_{ij} x_j \Rightarrow \frac{\partial f_i}{\partial x_j} = \underline{A_{ij}}$$

4. Gradient of Matrices



$y = \sigma(W\vec{x} + \vec{b})$, and we are searching for the "best" $W \Rightarrow$ derivative over matrix W

In general, compute the gradient of an $m \times n$ matrix A w.r.t. (with respect to) a $p \times q$ matrix B , the Jacobian]

would be $(m \times n) \times (p \times q)$ (4-dimensional tensor)

Case 1: Gradient of vectors w.r.t. matrices

Consider $\vec{f} = A\vec{x}$, with $\vec{f} \in \mathbb{R}^M$, $A \in \mathbb{R}^{M \times N}$, $\vec{x} \in \mathbb{R}^N$, compute $\frac{d\vec{f}}{dA}$

Answer: $\boxed{\frac{d\vec{f}}{dA}} \in \mathbb{R}^{\overset{M}{M} \times (M \times N)}$

$$\frac{d\vec{f}}{dA} = \begin{bmatrix} \frac{\partial f_1}{\partial A} \\ \vdots \\ \frac{\partial f_M}{\partial A} \end{bmatrix}, \quad \boxed{\frac{\partial f_i}{\partial A} \in \mathbb{R}^{1 \times (M \times N)}} \quad \downarrow$$

note that: $f_i = \sum_{j=1}^N A_{ij} x_j$, for $i=1, \dots, M$

Hence, $\checkmark \frac{\partial f_i}{\partial A_{ig}} = \checkmark \underline{x_g}$ $\uparrow$ $A_{ig} \cdot x_g$, $g=1, \dots, N$, therefore, one has

$$\frac{\partial f_i}{\partial A_i} = \underline{\vec{x}}^T \in \mathbb{R}^{1 \times (1 \times N)}$$

$$\underbrace{\left[\frac{\partial f_i}{\partial A_{i1}} \quad \dots \quad \frac{\partial f_i}{\partial A_{iN}} \right]}_{x_1 \quad \quad \quad x_N}$$

and $\frac{\partial f_i}{\partial A_{k \neq i}} = \underline{0}^T \in \mathbb{R}^{1 \times (1 \times N)}$
 $\uparrow$ the k -th row of A is not related to f_i

$$\Rightarrow \text{for } i \in [1, M], \quad \frac{\partial f_i}{\partial A} = \begin{bmatrix} \underline{0}^T \\ \vdots \\ \underline{0}^T \\ \underline{\vec{x}}^T \\ \vdots \\ \underline{0}^T \end{bmatrix} \in \mathbb{R}^{1 \times (M \times N)} \quad \begin{matrix} \uparrow \\ \text{i-th row} \end{matrix} \Rightarrow \frac{d\vec{f}}{dA} \text{ : aggregation of such matrices}$$

Case 2: Gradient of Matrices w.r.t. Matrices.

Consider a matrix $R \in \mathbb{R}^{M \times N}$, $\vec{f}: \mathbb{R}^{M \times N} \rightarrow \mathbb{R}^{N \times N}$, with

$$\vec{f}(R) = R^T R := K \in \mathbb{R}^{N \times N}, \quad \text{compute } \frac{dK}{dR}.$$

Answer: $\frac{dK}{dR} \in \mathbb{R}^{(N \times N) \times (M \times N)}$

note that $\frac{\partial K_{pq}}{\partial R} \in \mathbb{R}^{1 \times (M \times N)}$, for all $p, q=1, \dots, N$.
 $\leftarrow$ (p, q) th entry of K
 $\leftarrow$ p -th column of R

$$K_{pq} = \vec{r}_p^T \vec{r}_q = \sum_{i=1}^M R_{ip} \cdot R_{iq}$$

$$\Rightarrow \frac{\partial K_{pq}}{\partial R_{ij}} = \sum_{i=1}^M \frac{\partial}{\partial R_{ij}} R_{ip} R_{iq} = \delta_{pqij} = \begin{cases} R_{iq} & \text{if } j=p, p \neq q \\ R_{ip} & \text{if } j=q, p \neq q \\ 2R_{iq} & \text{if } j=p, p=q \Rightarrow R_{iq} \\ 0 & \text{otherwise} \end{cases}$$

Each entry of the desired gradient $\frac{dK}{dR} \in \mathbb{R}^{(N \times N) \times (M \times N)}$ is δ_{pqij}
 for $p, q, j=1, \dots, N$, $i=1, \dots, M$.

A few more useful identities for computing gradient.

$$\textcircled{1} \frac{\partial \vec{x}^T \vec{a}}{\partial \vec{x}} = \vec{a}^T = \frac{\partial \vec{a}^T \vec{x}}{\partial \vec{x}}; \textcircled{2} \frac{\partial \vec{a}^T \vec{x} \vec{b}}{\partial \vec{x}} = \vec{a} \vec{b}^T; \textcircled{3} \frac{\partial \vec{x}^T \vec{B} \vec{x}}{\partial \vec{x}} = \vec{x}^T (\vec{B} + \vec{B}^T)$$

$$\textcircled{4} \frac{\partial}{\partial \vec{S}} (\vec{x} - \vec{A} \vec{S})^T \vec{W} (\vec{x} - \vec{A} \vec{S}) = -2 (\vec{x} - \vec{A} \vec{S}) \vec{W} \vec{A} \text{ for symmetric } \vec{W} \checkmark$$

reference: The matrix Cookbook by Peterson and Pedersen 2012 for more useful properties

5. Taylor series

(1) single variable function: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f \in C^\infty$, the Taylor series f at x_0 is:

$$T_\infty(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

(to degree n : n instead of ∞)
 $x_0 = 0$, Maclaurin Series
 $\Rightarrow$ approximating original f
factorial

(2) multivariate Taylor series: Consider a function $f: \mathbb{R}^D \rightarrow \mathbb{R}$ which is smooth at $\vec{x}_0$

$$f(\vec{x}) = \sum_{k=0}^{\infty} \frac{D_{\vec{x}}^k f(\vec{x}_0)}{k!} (\vec{x} - \vec{x}_0)^k \quad (\text{to degree } n: n \text{ instead of } \infty)$$

$$\vec{s} = \vec{x} - \vec{x}_0 \quad \vec{s}^k \text{ definition?}$$

$$\vec{s}^2 = \vec{s} \otimes \vec{s} = \vec{s} \vec{s}^T \Rightarrow \vec{s}^2[i, j] = s[i] s[j]$$

$$\vec{s}^3 = \vec{s} \otimes \vec{s} \otimes \vec{s} = \vec{s}^3 \mathbb{E}[i, j, k] = s[i] s[j] s[k]$$

$$\Rightarrow D_{\vec{x}}^k f(\vec{x}_0) \vec{s}^k = \sum_{i_1=1}^D \dots \sum_{i_k=1}^D D_{\vec{x}}^k f(\vec{x}_0) [i_1, \dots, i_k] \cdot s[i_1] \dots s[i_k]$$

(3) linear Approximation of a function around a point x_0

$$f'(\vec{x}) = f(\vec{x}_0) + (\nabla_{\vec{x}} f)(\vec{x}_0) (\vec{x} - \vec{x}_0) \Rightarrow \text{Taylor series with order 1}$$

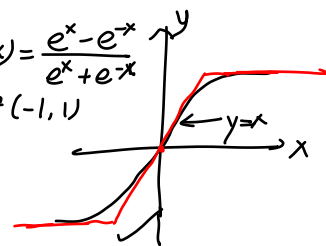
$\Rightarrow$ Approximation of the function around a point with linear curve.

Example. Consider a sigmoid-type function: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$\tanh'(x) = \frac{(e^x + e^{-x})(e^x - e^{-x}) - (e^x - e^{-x})(e^x + e^{-x})}{(e^x + e^{-x})^2} = 1 - \tanh^2(x)$$

around $x=0$

$$g_+(x) = \tanh(0) + x \tanh'(0) = 0 + x \cdot 1 = x$$



$$x \rightarrow +\infty, \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \approx \frac{e^x}{e^x} = 1$$

$$x \rightarrow -\infty, \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \approx \frac{-e^{-x}}{e^{-x}} = -1$$

summary: optimization $\rightarrow$ searching direction.

Taylor series $\downarrow$ linear approximation

derivative

- $\mathbb{R} \rightarrow \mathbb{R}$: derivative
- $\mathbb{R}^n \rightarrow \mathbb{R}$: Gradient vector w.r.t. a single variable
- $\mathbb{R}^m \rightarrow \mathbb{R}^E$: Grad. vec. w.r.t. vec
- $\mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{p \times q}$: Grad. w.r.t. mat